ESEMPI METODO DI GAUSS PIVOTING

ESERCIZIO

A= $\begin{pmatrix} \frac{3}{10} & \frac{1}{10} & 1 & \frac{1}{5} \\ \frac{1}{5} & 4 & -1 & 4 \\ \frac{1}{2} & \frac{41}{10} & 0 & 8 \\ \frac{1}{10} & -\frac{39}{10} & 2 & 9 \end{pmatrix}$ $b = \begin{pmatrix} \frac{1}{2} \\ 7 \\ 9 \\ 11 \end{pmatrix}$

Matrice completa:

$$\left(\begin{array}{cccc|c} \frac{3}{10} & \frac{1}{10} & 1 & \frac{1}{5} & \frac{1}{2} \\ \frac{1}{5} & 4 & -1 & 4 & 7 \\ \frac{1}{2} & \frac{41}{10} & 0 & 8 & 9 \\ \frac{1}{10} & -\frac{39}{10} & 2 & 9 & 11 \end{array}\right)$$

scambio riga 1 con riga 3

$$\left(\begin{array}{cccc|c} \frac{1}{2} & \frac{41}{10} & 0 & 8 & 9 \\ \frac{1}{5} & 4 & -1 & 4 & 7 \\ \frac{3}{10} & \frac{1}{10} & 1 & \frac{1}{5} & \frac{1}{2} \\ \frac{1}{10} & -\frac{39}{10} & 2 & 9 & 11 \end{array}\right)$$

I passo

$$\left(\begin{array}{cccc|c} \frac{1}{2} & \frac{41}{10} & 0 & 8 & 9 \\ 0 & \frac{59}{25} & -1 & \frac{4}{5} & \frac{17}{5} \\ 0 & -\frac{59}{25} & 1 & -\frac{23}{5} & -\frac{49}{10} \\ 0 & -\frac{118}{25} & 2 & \frac{37}{5} & \frac{46}{5} \end{array}\right)$$

scambio riga 2 con riga 4

$$\left(\begin{array}{cccc|c} \frac{1}{2} & \frac{41}{10} & 0 & 8 & 9 \\ 0 & -\frac{118}{25} & 2 & \frac{37}{5} & \frac{46}{5} \\ 0 & -\frac{59}{25} & 1 & -\frac{23}{5} & -\frac{49}{10} \\ 0 & \frac{59}{25} & -1 & \frac{4}{5} & \frac{17}{5} \end{array}\right)$$

II passo

$$\left(\begin{array}{cccc|c} \frac{1}{2} & \frac{41}{10} & 0 & 8 & 9 \\ 0 & -\frac{118}{25} & 2 & \frac{37}{5} & \frac{46}{5} \\ 0 & 0 & 0 & -\frac{83}{10} & -\frac{19}{2} \\ 0 & 0 & 0 & \frac{9}{2} & 8 \end{array}\right)$$

Il metodo di Gauss-Piv non può proseguire ! Difatti la matrice è singolare : $\det(A) = 0$.